Lab: Exploring Learning in Process Optimization

Objectives

> **Learning Goal:** Apply the Active Inference concept of learning to a concrete metallurgical scenario using digital twin lab techniques.

This is a **digital twin lab exercise** designed for materials scientists and engineers.

Part 1: System Definition (10 min)

Choose a metallurgical system to analyze. Suggested options:

- Fe-0.4wt%C steel undergoing austenitization and cooling
- Al-4wt%Cu alloy during precipitation aging
- Ti-6Al-4V produced by laser powder bed fusion
- Pure copper undergoing recrystallization after cold work

Write your response here

Part 2: Learning Analysis (25 min)

2A: Identify the Active Inference Components

Map your chosen system onto the Active Inference framework in the context of learning:

Component	Your System
System boundary (Markov blanket)	
Internal states	
External states	
Sensory states	
Active states	

Write your response here

2B: Learning Dynamics

Describe how learning operates in your system:

- What is the driving force (free energy gradient)?
- What is the timescale?
- What characterization technique would you use to observe it?

Write your response here

Part 3: Quantitative Exercise (20 min)

Perform a simple calculation or simulation related to learning in your system:

- If computational: use Python with PyCalphad, NumPy, or similar

- If analytical: use thermodynamic relations (Gibbs energy, diffusion equation, nucleation barrier)

Write your response here

Part 4: Reflection (10 min)

1. How did the Active Inference framework change your understanding of learning in this system?
2. What prediction error exists between the equilibrium model and the real behavior of your system?
3. How could you reduce this prediction error through better characterization or modeling?

Write your response here

Summary

Analysis Component	Key Finding
System analyzed	
Learning mechanism identified	
Driving force	
Main prediction error	
Recommended next characterization	

Write your response here